

Multiple Choice

1. **Answer:** A

Options: A) 20; B) 34; C) 40; D) $\sqrt{40}$

Note: Correct option: A - 20

2. **Answer:** D

Options: A) Domain and range remain the same; B) Domain is double; range remains the same; C) Domain is tripled; range is doubled; D) Domain is the same; range is tripled

Note: Correct option: D - Domain is the same; range is tripled

3. **Answer:** A

Options: A) 17 over 2; B) 34 over 4; C) 17 over 4; D) 19 over 2

Note: Correct option: A - 17 over 2

4. **Answer:** A

Options: A) 11; B) 22; C) 34; D) 48

Note: Correct option: A - 11

5. **Answer:** C

Options: A) 8; B) 15; C) 30; D) 45

Note: Correct option: C - 30

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '20'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '9'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3,750'.

Long Form Response

11. **Answer:** The areas of the four circles are π , 9π , 25π and 49π . The areas of the two black regions are π and $25\pi - 9\pi = 16\pi$, for a total black area of $\pi + 16\pi = 17\pi$. The areas of the two white regions are $9\pi - \pi = 8\pi$ and $49\pi - 25\pi = 24\pi$, for a total white area of $8\pi + 24\pi = 32\pi$. The ratio of the black area to the white area is $17\pi/32\pi = \frac{17}{32}$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** 6. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key